

Week 6 perceptron and linear regression

1. What is the perceptron? What are the steps in the algorithm(4)?

Perceptron is a classifier that is inspired by the brain. It takes an input and add in weights. Then it has an adder that does the dot-product on x and w . then it has an activation function that decides which class the input belongs to. The activation function in perceptron is a step function, meaning that if the product $>$ threshold then the class is 1, if the product $<$ threshold then the class is 0.

2. Why do we add the bias term to the perceptron?

We add a bias term (-1 or 1) to X_0 to fit the data better. it allows the perceptron to shift the decision boundary away from the origin and better fit the data. Without the bias term, the perceptron would be forced to pass through the origin (0,0) of the coordinate system, which may not always be the best fit for the data. Adding a bias term allows the decision boundary to be shifted up or down, left or right, depending on the data distribution and the desired classification outcome.

3. How do we initialize the weights? How do we update the weights in perceptron? What is the formula?

The weights can be set to 0 or is often given in the assignment. We update the weight depending on the output y . If y is different from t , then we have to change the weights. But if it is the same, then we do not change the weights. We iterate through each point and see if it was correctly classified, if it was not correctly classified we update the weight for each feature i with this formula:

$$W_i \leftarrow W_i + \text{Lr} * (y - t) * x_i$$

Lr is learning rate.

4. What is linear separability?

Linear separability is when we can separate the datapoints with a point, line, plane or hyperplane, depending on the number of features. All the points on one class fall on one side, and the other class on the other side.

5. What is a linear classifier?

A linear classifier is a classifier that separates the data into classes. A linear classifier always propose a linear decision boundary, whether the set is linear separable or not.

6. What is the goal in regression? What is the difference between classification and regression?

Regression assigns a numerical value to a point. The difference between classifier and regression is that classifier assign a class/category to a datapoint, while a regression assign a numerical value or real value.

7. What is linear regression?

Linear regression makes a line or linear function that is well suited for the training datapoints. It predicts a real value to a datapoint. Linear regression can only learn straight lines.

8. How can we tell whether a line is a good fit for the data?

We can tell if a line is good by looking at the loss. The most common function for calculating loss is Mean Square Error. The goal is to minimize this error to find the best fitting line. The function for MSE is

$\text{mean}((t-y)^2)$

9. What are the steps in linear regression?

The goal is to find a line that fits best to the datapoints. To find this we want to minimize the MSE. So we want to find the best weights that minimize MSE

$$f = \frac{1}{N} \sum_{j=1}^N (t_j - \sum_{i=0}^m w_i x_{ji})^2$$

To do this we use gradient descent.

We have the input datapoints X and the corresponding target values T . We add the bias and initialize the weights. The forward step is $Y = X@W$. Then we update the weights

$W -= \text{lr} * X.T(Y-T)$

Or $Q -= \text{lr} * X.T@ (Y-T)$

- Forward step:
 - $Y = X@W$
- Update step:
 - $W = W - \eta \nabla f$
 - $W -= \eta X.T(Y - T)$
 - (η , *eta*, is a learning rate)

10. What is inductive bias?

Inductive bias refers to the set of assumptions or constraints that a learning algorithm or model uses to make predictions or generalize from observed data to unseen instances. It represents the prior knowledge or assumptions encoded in the learning process, guiding the algorithm's search for the best hypothesis or model.

Inductive bias is necessary because learning algorithms face the challenge of generalizing from a limited set of observations to make predictions about new, unseen data. The choice of inductive bias influences the kind of patterns or relationships the model is more likely to learn or prefer.

In general, a learning algorithm in machine learning will carry certain assumptions regarding what the trained model should look like. Without assumptions, an algorithm would not be able to generalize from a finite set of observations.

Linear regression can only learn straight lines.

Perceptron can only learn linear decision boundary.